

# NCAA March Madness Prediction System

Project Update #2: Model Development, Barttorvik Integration & Backtest Results

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## 1 Overview

This report covers Weeks 2–4 (March 27 through April 10). We implemented nine prediction models spanning three tiers of complexity, expanded the feature set with Barttorvik continuous efficiency ratings, Dean Oliver’s Four Factors, and coach tournament experience features, and ran leave-one-tournament-out (LOTO) backtests across 11 seasons (2014–2025, excluding 2020). The headline result: integrating Barttorvik’s continuous adjusted efficiency metrics (AdjOE, AdjDE, NetRtg) broke through the ordinal rank ceiling identified in our initial experiments, yielding a multi-feature logistic regression with **Brier 0.159**—well below the Vegas benchmark of 0.175 and far exceeding our original target of  $<0.170$ .

## 2 Models

We implemented nine models across three tiers plus an optimized ensemble.

### 2.1 Tier 1: Baselines

**Seed Logistic.**  $P(A \text{ wins}) = \sigma(\beta_0 + \beta_1 \Delta s)$  where  $\Delta s$  is the seed difference. 0.199 Brier, consistent with literature.

**KenPom Logistic.** Same form on POM ordinal rank difference. 0.196 Brier. Only 0.003 better than seed, confirming the ordinal rank ceiling: integer ranks from 1 to 363 lose the continuous efficiency information that distinguishes close teams.

**Barttorvik Logistic.** Logistic regression on Barttorvik continuous net rating difference (AdjOE – AdjDE). **0.169 Brier.** This single feature, by being continuous rather than ordinal, outperforms every model from our initial experiments. This was the single most impactful change in the project.

**Elo.** Season-long Elo with  $K = 24$ , home-court advantage of 75 Elo points, 50% regression to the mean, and margin-of-victory adjustment ( $K \times \log(|\text{margin}| + 1) / \log 10$ ). 0.221 Brier. Despite tuning and MOV, Elo remains limited by its purely win/loss-based information.

**Efficiency Logistic.** Logistic regression on our self-computed net efficiency difference. 0.225 Brier.

## 2.2 Tier 2: Advanced

**XGBoost.** Gradient-boosted trees on a curated 24-feature set (down from 53), with stronger regularization: 150 estimators, max depth 3, min\_child\_weight 5, L1/L2 regularization. **0.172 Brier**, a dramatic improvement from the original 0.214. The improvement came from (1) adding Barttorvik continuous features, (2) removing 27 redundant correlated rank columns, and (3) stronger regularization to prevent overfitting on ~670 training games per fold.

**Multi-Feature Logistic.** Logistic regression with StandardScaler on 21 selected features including Barttorvik metrics, Four Factors, momentum, and coach tournament experience. **0.159 Brier**—our best single model, beating Vegas.

**Mixture of Experts (K=4).** Gating network routes matchups to specialized logistic experts by seed gap. 0.185 Brier (improved from 0.209 via Barttorvik features).

**Transformer (4-layer → 2-layer, pretrained).** We redesigned the Transformer with a smaller architecture (d\_model=32, 2 layers, 2 heads, ~8K parameters) and added a three-phase training pipeline: (1) pretrain encoder on ~5K regular-season game pairs per season with causal masking, (2) freeze encoder and train matchup MLP on tournament data, (3) unfreeze with  $0.1 \times$  learning rate for joint fine-tuning. **0.196 Brier**, improved from 0.245. Pretraining on regular-season data was essential—without it, the model could not learn meaningful representations from only 736 tournament games.

## 2.3 Tier 3: Meta

**Weighted Ensemble.** Optimized weighted average of the top 4 models (Multi-Feature Logistic, Barttorvik Logistic, Seed Logistic, XGBoost), with weights optimized per LOTO fold via inner leave-one-season-out cross-validation to minimize Brier score. 0.161 Brier. The optimizer consistently assigned 80–95% weight to Multi-Feature Logistic, indicating that the logistic model already captures most available signal. The ensemble provides marginal improvement on some seasons but does not consistently beat the best single model.

## 3 Feature Improvements

### 3.1 Barttorvik Continuous Efficiency Ratings

This was the breakthrough feature of the project. We built an automated scraping pipeline that fetches Barttorvik T-Rank data for all D1 teams across seasons 2014–2025, handles JavaScript browser verification, parses HTML tables, and fuzzy-matches team names to Kaggle TeamIDs using the MTeamSpellings.csv mapping (with rapidfuzz for unmatched names). Results are cached to data/external/barttorvik\_{season}.csv.

Five new matchup features were added:

- **bart\_net\_diff:** Net rating difference ( $\text{AdjOE} - \text{AdjDE}$ ). The single most predictive feature.
- **bart\_adjoe\_diff:** Adjusted offensive efficiency difference.
- **bart\_adjde\_diff:** Adjusted defensive efficiency difference.
- **bart\_barthag\_diff:** Barthag (win probability proxy) difference.
- **bart\_tempo\_diff:** Adjusted tempo difference.

The key insight is that Barttorvik’s AdjOE/AdjDE are *continuous* efficiency margins (points per 100 possessions), not ordinal ranks. The difference between the #1 and #5 team might be 8 points of net rating, while the difference between #50 and #55 might be 0.3 points. Ordinal

ranks cannot distinguish these cases; continuous ratings can. This single change reduced our best Brier score from 0.196 to 0.169 (single feature) and 0.159 (multi-feature).

### 3.2 Four Factors

Dean Oliver’s Four Factors from Update #1 remain in the feature set. They contribute modestly when Barttorvik features are present, as Barttorvik’s adjusted metrics already incorporate similar information.

### 3.3 Coach Tournament Experience

We added two features from `MTeamCoaches.csv`: the number of prior NCAA tournament appearances for each team’s head coach, and total coaching seasons. The impact was negligible ( $<0.001$  Brier), suggesting that coach experience is already captured by team quality metrics.

### 3.4 Feature Selection for XGBoost

We curated the XGBoost feature set from 53 to 24 features by: (1) dropping individual `rank A/rank B` columns (keeping only diffs), reducing 27 rating columns to 9; (2) keeping only 3 of 9 rating systems (POM, SAG, RPI) since the others are  $>0.95$  correlated; (3) adding all 5 Barttorvik features. Combined with stronger regularization, this fixed the XGBoost overfitting problem ( $0.214 \rightarrow 0.172$ ).

## 4 Backtest Results

Table 1 reports the full LOTO results.

Table 1: Leave-one-tournament-out Brier scores (2014–2025, 11 folds). Lower is better. Benchmarks: coin flip = 0.250, seed baseline  $\approx 0.200$ , Vegas  $\approx 0.175$ .

| Model                    | Mean Brier   | Std   | Best / Worst Season |
|--------------------------|--------------|-------|---------------------|
| Multi-Feature Logistic   | <b>0.159</b> | 0.020 | 2025 / 2023         |
| Weighted Ensemble        | 0.161        | 0.021 | 2025 / 2014         |
| Barttorvik Logistic      | 0.169        | 0.022 | 2025 / 2018         |
| XGBoost (regularized)    | 0.172        | 0.025 | 2025 / 2014         |
| Mixture of Experts (K=4) | 0.185        | 0.023 | 2025 / 2023         |
| Transformer (pretrained) | 0.196        | 0.024 | 2025 / 2014         |
| KenPom Logistic          | 0.196        | 0.017 | 2025 / 2021         |
| Seed Logistic            | 0.199        | 0.021 | 2025 / 2021         |
| Elo (MOV-adjusted)       | 0.221        | 0.029 | 2014 / 2021         |
| Efficiency Logistic      | 0.225        | 0.016 | 2015 / 2022         |

Several patterns emerge. First, the top four models all use Barttorvik continuous features, confirming that this was the decisive improvement. Second, the Weighted Ensemble (0.161) barely beats the best single model (0.159), because the optimizer assigns 80–95% weight to Multi-Feature Logistic. Third, the Transformer improved dramatically from 0.245 to 0.196 with pretraining, but still cannot compete with the statistical models that use expert-constructed features like Barttorvik ratings.

Year-to-year variance remains large ( $\text{std} \approx 0.02$ ), comparable to model gaps. 2025 was the best season for every model (unusually chalky tournament), while 2021 and 2014 were consistently the hardest.

## 5 What Worked and What Didn't

### 5.1 Breaking the ordinal rank ceiling

This was the project's central breakthrough. Our initial experiments hit a wall at Brier 0.196 because every feature was an integer rank. Barttorvik's continuous AdjOE/AdjDE/NetRtg immediately broke through this ceiling. A single continuous feature (net rating difference) at 0.169 outperforms 53 ordinal features fed to XGBoost (0.214). The lesson is clear: *feature quality dominates model complexity*.

### 5.2 XGBoost: less is more

XGBoost improved from 0.214 to 0.172 through three changes: (1) adding 5 Barttorvik features, (2) removing 29 redundant rank columns, (3) stronger regularization. The regularization changes alone (depth 4→3, 300→150 trees, added L1/L2) would not have helped without the feature improvements. XGBoost needs good features even more than simple models do, because it has more capacity to overfit on noise.

### 5.3 Transformer pretraining: necessary but not sufficient

Pretraining on ~5K regular-season game pairs per season reduced the Transformer from 0.245 to 0.196. The three-phase training pipeline (pretrain, freeze+finetune, unfreeze with low LR) was essential. However, 0.196 only matches KenPom Logistic. The Transformer captures temporal dynamics (momentum, trajectory) but cannot access the cross-team adjusted efficiency information that Barttorvik provides. A future direction is to include Barttorvik features in the game embedding.

### 5.4 Ensemble: diminishing returns

The optimized ensemble (0.161) barely improves on Multi-Feature Logistic (0.159). The weight optimizer consistently puts >80% on the logistic model, meaning the other models contribute little unique signal once Barttorvik features are included. This suggests we are near the information ceiling of our feature set.

### 5.5 Calibration and coach features: negligible impact

Isotonic calibration on XGBoost slightly *worsened* performance (0.172 → 0.174), suggesting the model is already well-calibrated. Coach tournament experience features contributed <0.001 Brier improvement, indicating that team quality metrics already subsume coaching effects.

## 6 Next Steps

Three directions for the final phase (Weeks 4–6):

1. **Hybrid Transformer.** Inject Barttorvik features into the Transformer's game embeddings or add them as a separate static branch concatenated with the CLS token. This could combine the Transformer's temporal modeling with the logistic model's feature advantage.
2. **Market calibration.** Integrate historical betting odds via the `MarketCalibratedEnsemble` (already implemented but unused) for Bayesian fusion of model predictions with market-implied probabilities.
3. **2026 tournament prediction.** Generate bracket predictions for the live 2026 tournament and submit to the Kaggle March Machine Learning Mania competition.

Our current best (0.159) is competitive with top Kaggle entries. The remaining gap to the theoretical optimum ( $\sim 0.140$ – $0.150$  estimated from betting market efficiency) likely requires player-level features or proprietary data that we do not have access to.

## References

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